

Distributed Audio DSP Middleware — Milestone 3 Update

Paul Bonargo Jr. · M.S. Computer Science, Rochester Institute of Technology · CSCI-788 Capstone · Advisor: Prof. Abeer Ahmad · July 16, 2026

Status: Milestone 3 service foundation complete. The remote node is now a native C++/JUCE application processing the live audio stream, and the host plugin and node communicate over a reliable TCP control channel with automatic reconnection. Together these form the foundation that remote plugin hosting requires; hosting itself is designed, in progress, and re-scoped to Milestone 3+ (below).

What was completed this milestone

- Native C++ processing node. The Milestone 2 Python node was replaced by a standalone C++/JUCE application: a real-time processing engine (receive → process → return over dual UDP ports) speaking the project's custom 36-byte v2 wire protocol, with a status window reporting live packet counts. It currently runs in echo (pass-through) mode — the exact engine that hosted plugins will slot into.
- TCP control plane. A reliable control channel (length-framed JSON over TCP) now runs alongside the UDP audio path: a handshake carrying the host's sample rate and block size, plugin-list delivery, and plugin-select / parameter-change message types. Control messages must never be dropped, so control rides TCP while audio deliberately stays on UDP.
- Automatic reconnection. If the node disappears, the plugin detects the lost control connection and re-establishes it within $\approx 1\text{--}2$ seconds of the node returning — no restart required on either side.
- Runtime node selection. The audio destination and control connection now retarget at runtime; switching processing nodes no longer requires rebuilding the plugin.
- Validation. Scripted loopback (≈ 750 packets over 2 s, all validated against the v2 header); live Ableton playback sustained at the stream's inherent 375 packets/s; fault test — killing the node mid-playback produced no audible dropout (the dry-signal fallback engaged) and relaunching it auto-recovered the stream; TCP handshake and plugin-list delivery verified end to end.

Changes to scope

- Milestone split (deliberate). Milestone 3 was originally framed to also host real third-party VST3 plugins on the node with a live parameter-control interface. Given a compressed timeline (full-time co-op, 5 interviews, plus concurrent report deliverables), the milestone was scoped to the node application and control plane, and hosting moved intact — with a complete implementation plan for Milestone 3+. This mirrors the Milestone 2 split: the highest-variance work moves whole, and the prerequisites it depends on are locked in first.
- Foundation-first ordering. The control plane built this milestone is the seed of the target architecture's coordination layer (node scheduling, fault-tolerant failover), so the work advances the thesis independently of hosting.

Updated plan

- Milestone 3+ — Remote VST3 hosting + live parameter control. The node hosts real, selectable third-party VST3 plugins from an explicit registry; the plugin's interface gains a parameter panel that scales with the chosen plugin; hosted-plugin latency folds into the delay-compensation budget; validated with a two-machine LAN measurement (advisor demo).
- Milestone 3+ stretch — Offline freeze/flatten. Capture a pass, process it losslessly on the node over a reliable TCP lane, and cache the result locally for network-free playback — the loss-immune complement to the live tier.
- Beyond — Cross-network operation (VPN overlay first, NAT traversal later); cloud tier with license-aware offloading; fault tolerance on the control plane with node failover.

Milestone 3 deliverables (plan, implementation, validation captures) are committed to the project repository and filed under Documents directory. Known, documented limitations this milestone: node processing is echo/pass-through (real plugin hosting is Milestone 3+); validation testing was primarily loopback-based (the LAN measurement moves with hosting); linear-playback timeline assumption carried from Milestone 2.